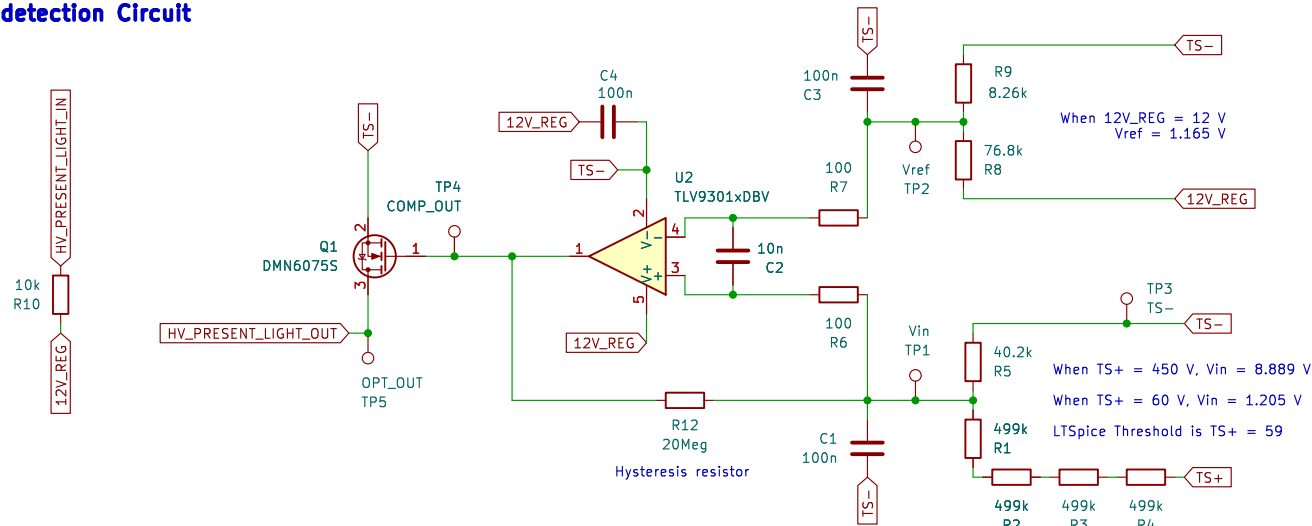


HV detection Circuit

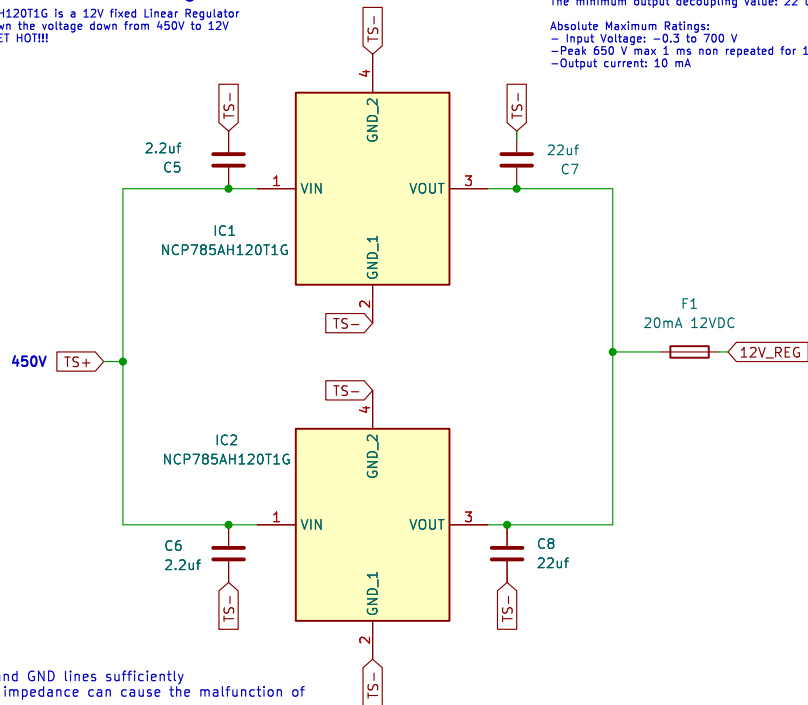


450V to 12V Linear Regulators

The NCP785AH120T1G is a 12V fixed Linear Regulator and steps down the voltage down from 450V to 12V THESE CAN GET HOT!!!

The minimum input decoupling value: 2.2 uF
 The minimum output decoupling value: 22 uF

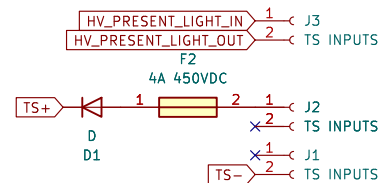
Absolute Maximum Ratings:
 - Input Voltage: -0.3 to 700 V
 - Peak 650 V max 1 ms non repeated for 1 s
 - Output current: 10 mA



Make VIN and GND lines sufficiently wide. High impedance can cause the malfunction of regulator.

Set external components, especially the output capacitor, as close as possible.

INPUTS & OUTPUTS



Branding Block (always include)

Last updated
26-06-2023

Standard
Branding Block
Edit in Layout

U3

Standard
Branding Block
Edit in Layout

U4

Oscar Rudman
NU Racing

Sheet: /
 File: AIL_V2.2.kicad_sch

Title: AIL_V2.2

Size: A4 Date: 2025-03-05

KiCad E.D.A. 9.0.0



Rev: 001

Id: 1/1